CarbonLedger Live

Tamper-Proof Carbon Credit System

CarbonLedger is a terminal-first teaching system for recording eco-friendly actions, awarding green points, detecting duplicate claims, verifying ledger integrity, ranking contributors, and comparing green-fund allocation strategies.

System overview

Each ledger record is freshly hashed from its current fields. Leaf hashes are combined through a Merkle tree until they produce a root hash. If a record changes, its leaf and every parent on the path to the root change. Comparing the current root with a trusted root identifies that the ledger is no longer intact; walking mismatched branches locates the affected record in logarithmic node visits.

An AVL tree keeps users or departments searchable and sorted with logarithmic insert and lookup. A trie supports prefix-based autocomplete for known eco-actions. A chained hash table rejects duplicate user-plus-action claims in constant average time. A max-heap produces top contributors without sorting the entire registry. The allocation module compares dynamic programming, greedy, and branch-and-bound approaches to the 0/1 knapsack problem.

Complexity profile

Operation	Structure or technique	Complexity
Insert user	AVL tree	O(log n)
Duplicate check	Hash table	O(1) average
Action lookup	Trie	O(L)
Leaderboard top-K	Max-heap	O(n) build + O(k log n)
Verify ledger	Merkle tree	O(n)
Locate tampered record	Merkle tree	O(log n)
Budget allocation	Dynamic programming	O(n · W)

Limitations and future scope

The current C implementation is single-user and uses fixed-capacity arrays with raw binary persistence. The teaching hash is djb2 rather than a cryptographic hash, and the saved binary file is not encrypted or signed. Future work could add digital signatures, encrypted storage, multi-user access, and a production API.

Project stack

C99, GCC, standard library only, Merkle tree, AVL tree, trie, chained hashing, max-heap, dynamic programming, greedy allocation, and branch-and-bound search.

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